

NAME: _Tovi_Ding_ SECTION DAY/TIME: __103 Tue Thu 8-10am__
 GSI: ____Idris_____ LAB PARTNERS: ____Ella_____

Equipotential lines and electric fields

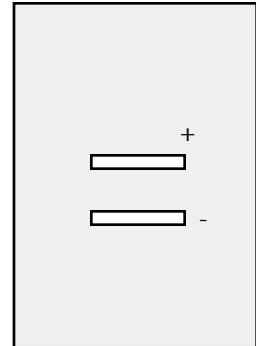
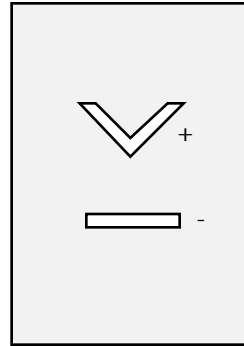
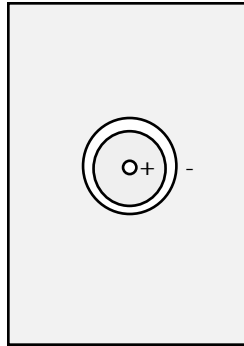
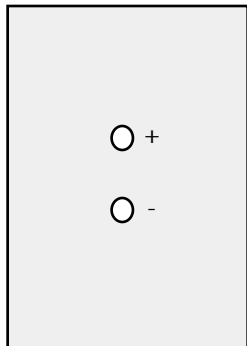
Introduction

This lab gets at one of the most difficult concepts in the course: electric potential, and its relationship to electric fields and potential energy. The lab activities and associated questions can help you get a real-world feel for these concepts and their conceptual underpinnings. The hardest thing about potential, however, is to see how all this fits together.

The activities below are designed to take only a portion of this period. We will take the rest of the time to continue with discussion section activities.

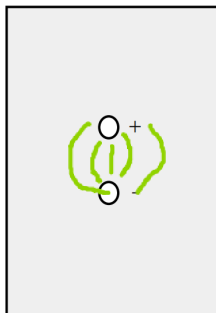
Questions

- In these pictures, the two conducting regions (marked in white) carry equal and opposite charges. What will the *equipotential lines* look like in each case? Remember, two points have the same potential if a charge would have the same potential energy at either point. Sketch your predictions here using dotted lines.



Now pick *two* of the four configurations, and experimentally sketch the equipotential lines, using the procedure on the next page. You should do one; your partner the other. To check your other two predictions, you can look at the work done by other lab tables.

we're doing 1 and 3

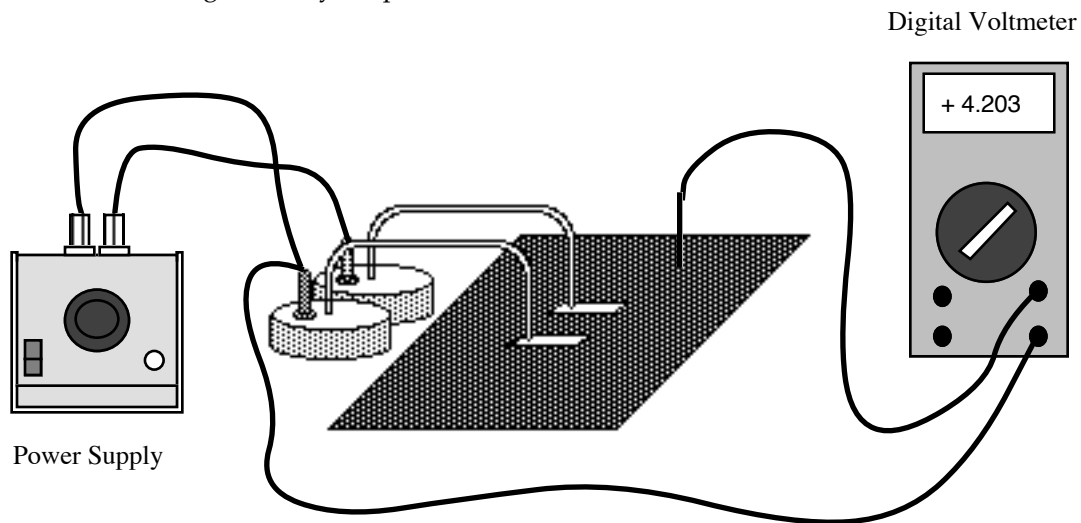


Before starting the experiment, make sure the equipment is working properly, and answer a brief question.

- ◆ Layer a piece of paper on the bottom, then a piece of carbon paper (dark side down), then a piece of the teledeltos paper with conducting paint regions on top. Tape two corners down so you can keep them aligned but still lift up to write on the bottom paper.
 - ◆ The equipment should already be set up as drawn on the next page, with the power supply set to 5 volts. So, the power supply enforces a 5 volt potential difference between the two metallic regions on your paper. Touch the voltage probe to one metallic region, and then to the other. It should register 0 V and then plus or minus 5 V (or vice versa).
2. Prediction: *Within* a metallic region, is the potential the same everywhere, or does it vary point by point? *Explain* why; don't just quote a result. (Note, within the metallic region means within the actual metal, not inside a cavity or region surrounded by metal.)

Prediction: the potential should be the same everywhere because metal's conductive

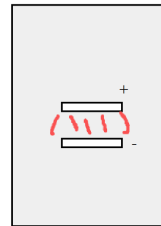
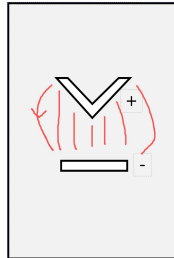
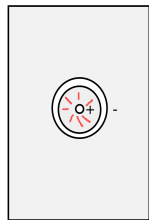
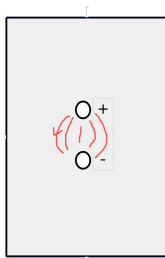
- ◆ Now, test *two* of the predictions you made earlier with the voltage probe (the “free” wire sticking out of the digital voltmeter). Use the following procedure. Remember: you should test one configuration, your partner the other.



Procedure for “sketching” the equipotential lines.

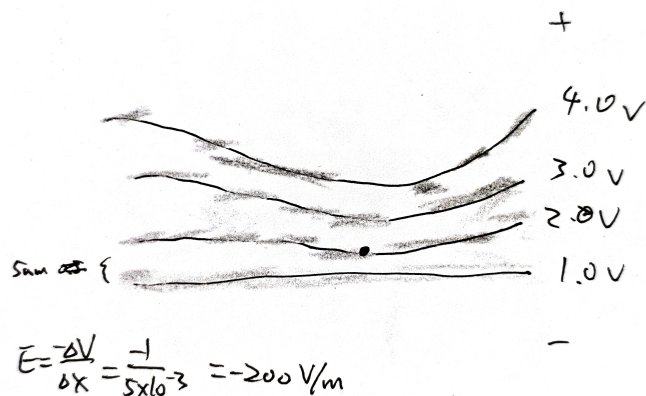
- ◆ Using the voltage probe, find a place on the paper where the potential is 1.0 volt. If it's 1.03 V or 0.98 V, that's fine; just get close. Mark that point by lightly rubbing the probe on the top layer. Check the bottom layer to make sure the mark was transferred by the carbon paper. Then, find another 1-volt point, about a centimeter from the first one. Mark it. And so on. By using symmetry and your above prediction, you may be able to save yourself some work.

- ◆ On the bottom sheet of paper, connect the dots. This curve is an equipotential line; every point along the curve has potential 1 volt.
 - ◆ Now make the equipotential lines for 2 V, 3 V, and 4 V, again using light pressure to mark the locations and connecting the dots on the bottom sheet of paper. Work pretty fast; it's more important to think about what these lines mean than it is to draw them perfectly.
 - ◆ If you've made any major errors in your predictions, please correct them now.
3. Using solid lines, add sketches of the electric field for each configuration on p. 1. Explain here how you know how to draw the fields.



simply connect the positive and negative charges

- 4.
- ◆ Estimate the electric field at a point you select between the two conductors on your plot. Record your data here, and explain your measurement. Hint: $E_x = -dV/dx$.
- because E is at the direction of change of V, it is always perpendicular to the equal potential line

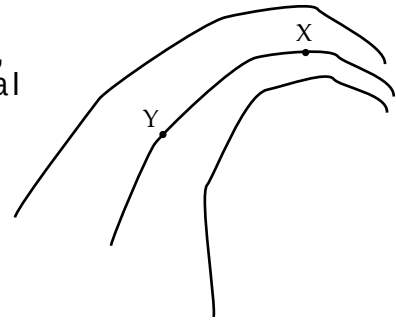


Please lightly erase any stray marks on your conducting paper, so that the next lab group gets a fresh start.

5. (Just a question, not an experiment.) In figure 5, at which of these two points, X or Y, is the electric field stronger? How do you know?

At X, because for the same change of V , the distance between the equal potential lines are smaller. Thus $E = dV/dx$ is larger at X

FIGURE 5
Segments of equipotential lines

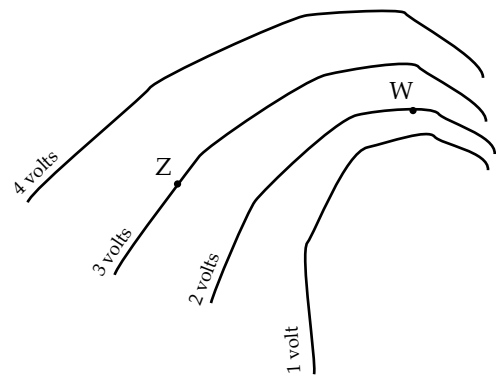


6. (a) In question 5, you compared the electric field at two points on the same equipotential line. Now consider two points on different equipotential lines. In figure 6, where is the field stronger—point W or point Z? How do you know?

W.

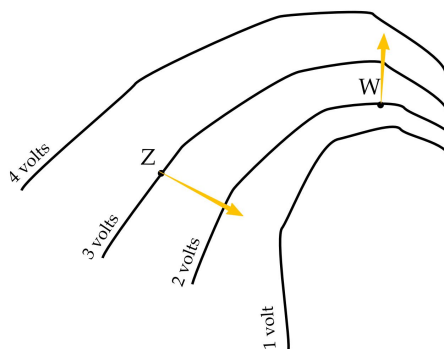
Same reason as Q5. The equal potential lines are closer at W. So for a change in V the change in x is smaller at W. Thus $E = dV/dx$ is larger at W than Z

FIGURE 6
Segments of equipotential lines

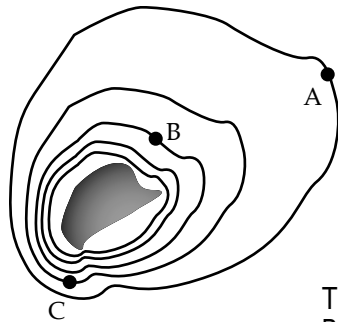


- (b) Sketch on figure 6 the direction in which a positive charge placed at point Z would move. Sketch the direction in which a negative charge placed at point W would move.

FIGURE 6
Segments of equipotential lines



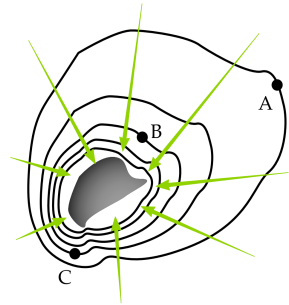
7.



The gray charge distribution shown generates an electric field corresponding to the following equipotential surfaces. The potentials at points A and B are $V_A = 3.0 \text{ V}$ and $V_B = 1.0 \text{ V}$.

- (a) On this diagram, sketch some of the electric field lines resulting from the charge distribution. Is the charge distribution positive or negative? (Yes, you have enough information to tell.)

The charge distribution is negative
Because the field lines are pointing inwards,
analogous to how electric field points to negative charge q



- (b) Where is the electric field strongest? Explain.

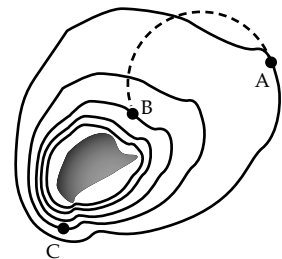
The electric field is the strongest at C because the equipotential line is the most dense.

- (c) How much work would it take to move a $Q = 0.50 \text{ C}$ point charge along a straight line from B to A?

$$W = -q \Delta V = -0.50 \text{ C} \cdot (3.0 \text{ V} - 1.0 \text{ V}) = -1 \text{ J}$$

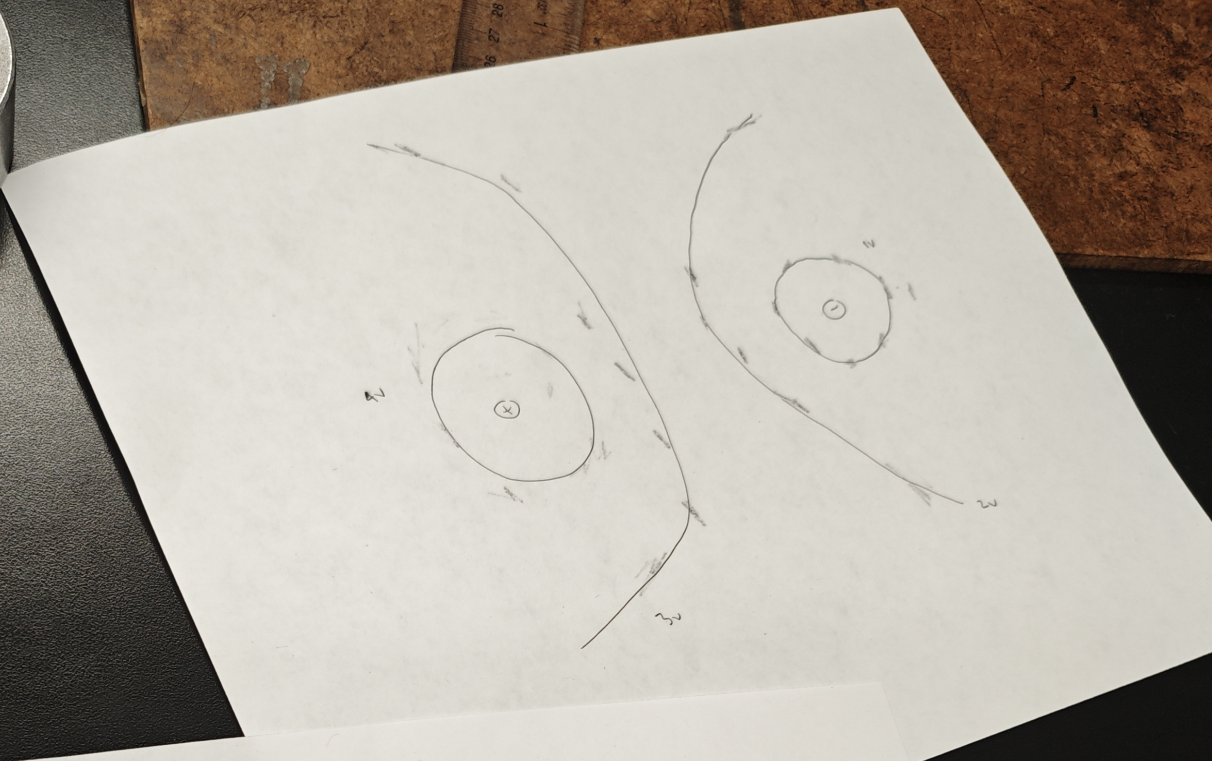
- (d) Now consider a semicircular path from B to A. To move the $Q = 0.50 \text{ C}$ charge along this path, would it take more work, less work, or the same work, as compared to part (c)? Explain.

The same. Because just like gravitational fields, electric field is also conservative.

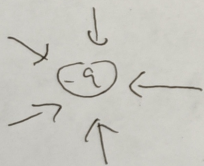


- (e) Which takes more work: Moving charge Q from point C to point A, or moving it from point B to point A? Justify your answer.

B to A, because $\Delta V = 3 - 1 = 2 \text{ V}$, which is larger than that of C to A, which is $3 - 2 = 1 \text{ V}$



③ ∇



$$E = \frac{\Delta V}{\Delta x} = \frac{1}{5 \times 10^{-3}} = -200 \text{ V/m}$$